Ganesh Gupta

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

Experience

Oracle | *Software Engineer, AI Platform* Jun 2025 – Present

- Built a **distributed executor** in **Java** and **Python** exposing typed **REST APIs** that dispatch workloads to **Docker** or serverless compute, holding **sub-200ms p99** across **millions of daily transactions**.
- Designed a content-addressed **S3** artifact store with memoization and **Redis** checkpointing so interrupted workflows resume from the last committed step, at **99%+ uptime**.
- Built **Elasticsearch**-backed search and retrieval services with knowledge-graph enrichment, secured with **OAuth/RBAC**.
- Shipped containerized services to **Kubernetes** through **GitHub Actions CI/CD**, with **SSE** streaming that cut redundant API calls **90%**.

QureAI | *AI Engineer* Mar 2026 – Present

- Cut **p99 latency from 420ms to 140ms** across **50K+ daily requests** by moving synchronous services to **async processing** with horizontal **autoscaling**.
- Built **PySpark** jobs and **Airflow** DAGs for large-scale data processing, orchestrating multi-stage pipelines with dependency management and failure alerting.
- Built **CI/CD** pipelines with automated quality gates and **rollback** for safe production releases.

Nomura Research Institute Financial Technologies | *Software Engineer* Sep 2019 – Aug 2023

- Architected **25+ Java Spring Boot microservices** with **REST APIs** for financial transaction processing, serving **1M+ users** at **99.9% uptime** on **AWS (EC2, RDS, SQS)**.
- Cut nightly aggregation **81% (8 minutes to 90 seconds)** by redesigning **SQL and PL/SQL** with **indexed views** and **partitioning** to remove row-level lock contention.
- Cut production escalations **60%** in one quarter by standardizing retry, timeout, and failure contracts across 25+ services and driving **root cause analysis** and post-incident reviews.
- Led **architecture and code reviews** and mentored engineers on object-oriented design, **design patterns**, and testable, maintainable backend code.

Projects

Distributed NoSQL Database `github.com/ganeshhgupta/distributed-kv-store`

- **Fault-tolerant** distributed database that confirms writes only after a **majority of nodes** store them, recovers crashed nodes from a **write-ahead log**, and resyncs replicas via **leader-follower replication**.

Watch-your-LLM `github.com/ganeshhgupta/watch-your-llm`

- Observability platform that traces model calls, latency, and token usage, with a **FastAPI** backend, **PostgreSQL**, and a **React/TypeScript** dashboard.

RootCause AI `github.com/ganeshhgupta/rootcause-ai`

- **LangGraph** multi-agent system that triages production alerts, routes each to a specialized sub-agent with **tool use**, and writes a structured **root-cause brief**.

Education

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science* Aug 2016 – Jul 2020

Skills

Languages: Java, Python, SQL, TypeScript, C++

Java and Backend: Spring Boot, Spring MVC, REST APIs, GraphQL, Microservices, JUnit, Mockito, OOP, Design Patterns

Databases: PostgreSQL, Oracle DB, MongoDB, DynamoDB, Redis, Elasticsearch

Cloud and Data: AWS, Docker, Kubernetes, GitHub Actions CI/CD, Airflow, PySpark, Kafka

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer